Reliability-Aware Spatiotemporal Consistency Regulation for Satellite Video Tracking

Xiaowen Zhang, Haijiang Sun, Hanqing Sun, Hongliang Chang, Xilai Wei, and Chao Ma

***Abstract*—The satellite video single object tracking (SVSOT) remains challenging because low image quality and cluttered backgrounds yield unreliable target evidence. Although recent trackers have improved robustness by enhancing feature representations and cross-frame interactions, they lack regulation of the reliability of evidence used for recursive feature updates. As single object tracking (SOT) can be seen as a recursive process, once unreliable evidence is absorbed into the target state, it may bias subsequent associations, accumulate spatiotemporal inconsistency, and eventually cause tracking drift or failure. To tackle these issues, we propose SCRTrack, a spatiotemporal consistency regulation framework that regulates the formation of target evidence and its propagation and maintenance across frames. Specifically, a multilayer feature formation (MFF) module is proposed to inject shallow structural details into deep semantic representations, enhancing spatial discriminability while suppressing clutter-induced interference. A reliability-guided context propagation (RCP) module is further designed to estimate interframe correspondence reliability and uses it to modulate temporal state updates, reducing the impact of ambiguous associations on state evolution. In addition, a motion consistency preservation (MCP) mechanism is introduced to monitor motion-consistent response displacements under occlusion and similar-object interference, maintaining target identity when appearance-based evidence becomes unreliable. Extensive experiments on SatSOT, SV248S, and OOTB demonstrate that SCRTrack achieves the state-of-the-art performance and exhibits strong robustness across challenging satellite tracking scenarios.**

***Index Terms*—Feature regulation, satellite video tracking, single object tracking (SOT), spatiotemporal consistency.**

I. Introduction

SATELLITE video surveillance enables wide-area and persistent monitoring from a long-distance perspective, making it an essential technology for traffic management, military reconnaissance, and disaster relief [1], [2], [3]. As a fundamental task in satellite video analysis, the satellite video single object tracking (SVSOT) aims to continuously localize a designated target across frames. Despite its importance, the SVSOT remains challenging due to degraded target observability. As illustrated in Fig. 1(a), the long imaging distance and large field of view result in low spatial resolution, where the target occupies only a few pixels and exhibits weak appearance cues, further blurring the distinction between target and surroundings. Meanwhile, target rotation and environmental fluctuations introduce temporal ambiguity, making interframe target association unreliable. Therefore, the challenge of SVSOT is not only weak per-frame appearances but also the inconsistent scenario variations across frames.

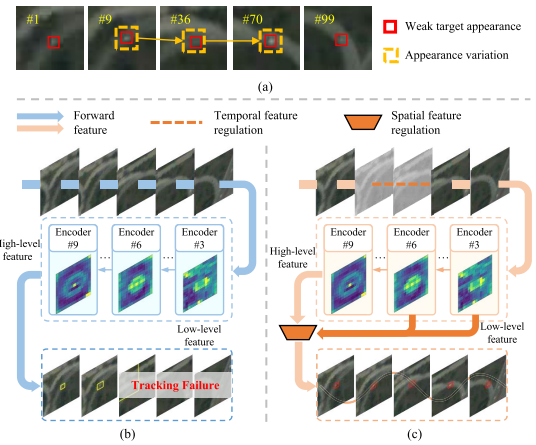


Fig. 1. (a) SVSOT suffers from unreliable observations caused by weak target appearance, cluttered backgrounds, and occlusion. (b) Existing methods recursively update target representations without reliability regulation, and these trackers mainly rely on high-level features, leaving low-level cues unregulated. (c) In contrast, SCRTrack calibrates spatial representations and regulates temporal features to preserve reliable target evolution.

Since tracking is a recursive state estimation process, target evidence in each frame is incorporated through recurrent interactions with historical states [4], [5], [6], [7]. This recursive absorption of unreliable evidence may cause spatial errors to be propagated temporally, leading to progressive spatiotemporal inconsistency, tracking drift, and identity ambiguity, especially in degraded satellite videos with occlusion or similar-object interference, as shown in Fig. 1(b). Hence, the key difficulty of robust SVSOT lies not merely in extracting discriminative features from degraded frames but also in regulating whether and how newly observed features should participate in target state update.

To improve tracking robustness, recent satellite trackers have evolved from framewise spatial modeling [8], [9], [10], [11] toward spatiotemporal learning [12], [13], [14] that exploits cross-frame continuity and historical context.

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However, most methods primarily focus on enriching spatial representations or aggregating more temporal context, while paying less attention to the reliability of the evidence admitted into the recursive target state. For spatial evidence formation, multilevel fusion [15] can improve localization by exploiting fine-grained details, but direct aggregation may also mix discriminative target cues with clutter-sensitive background responses, resulting in biased target representation. For temporal evidence propagation, template updating [16] and memory mechanisms [14] strengthen cross-frame dependency modeling, yet they rarely constrain whether the propagated historical context is supported by reliable interframe correspondences. As a result, ambiguous associations may persist in the target state and gradually amplify tracking errors. Furthermore, while some motion-based trackers [17], [18] maintain identity consistency without relying on visual cues, their reliance on single-motion constraints brings challenges in complex scenarios involving simultaneous occlusion and distractor interference. In general, the existing methods still lack explicit regulation over how target evidence is formed, propagated, and maintained during recursive tracking.

To address the above limitations, we propose SCRTrack, a spatiotemporal consistency regulation framework for robust SVSOT, as illustrated in Fig. 1(c). Instead of solely enhancing feature extraction or temporal modeling, SCRTrack formulates them as coupled regulation operations. Specifically, SCRTrack regulates target evidence from three tightly related aspects, spatial evidence formation, temporal evidence propagation, and identity-preserving state maintenance. First, a multilayer feature formation (MFF) module is introduced to integrate fine-grained structural cues into deep semantic representations, forming discriminative target representations. Second, a reliability-guided context propagation (RCP) module is designed to estimate the reliability of interframe correspondences and use it to modulate temporal state evolution, preventing ambiguous associations from dominating the update process. Third, a motion consistency preservation (MCP) mechanism is proposed to monitor motion-consistent response displacements and predict target trajectories, which alleviates identity switching under occlusion and similar-object interference. By jointly regulating how target evidence is formed, propagated, and maintained, the SCRTrack enables more stable tracking in satellite videos. In summary, this work makes the following main contributions.

1) We identify unreliable evidence absorption during recursive target state update as a key cause of spatiotemporal inconsistency accumulation in SVSOT and cast robust satellite tracking from the perspective of reliability-aware spatiotemporal regulation.
2) We propose SCRTrack, a unified spatiotemporal consistency regulation framework that controls how target evidence is formed, propagated, and maintained, thereby reducing the impact of degraded observations, ambiguous correspondences, and distractor interference.
3) We design three components. MFF forms discriminative features with detailed structures and semantic representations, RCP propagates reliable target evidences according to correspondence reliability, and MCP maintains target identity by monitoring motion-consistent response displacements.
4) Extensive experiments on SatSOT, SV248S, and OOTB demonstrate that SCRTrack achieves the state-of-the-art performance and exhibits strong robustness in challenging satellite tracking scenarios.

II. Related Work

A. Satellite Video SOT

The SVSOT aims to continuously localize a target in satellite video sequences. With the development of deep learning, the Siamese-based trackers have become a mainstream paradigm for SVSOT. HRSiam [15] directly aggregates multiresolution features to strengthen target representation, whereas SiamMDM [16] dynamically models target appearance and motion across frames. SiamS^{2}F [19] further exploits multidimensional target cues for spatial–spectral feature enhancement. The motion-based methods have also been developed to alleviate temporal instability, including KF-based motion compensation in CFME [17] and trajectory regression in LMANet [20]. Recently, the transformer-based satellite trackers have explored spatiotemporal fusion in STAR [12], high-order temporal relations in HRLT [21], matching–propagation–prediction pipelines in TransMPP [22], temporal memory in MemTrack [14], and hybrid offline–online adaptation in HLSOT [23]. Recently, MTTrack [24] introduced Mamba into satellite video tracking to model intergroup dependencies and extract discriminative visual representations. Unlike its representation-oriented Mamba modeling, the SCRTrack uses tokenwise correspondence reliability to regulate recurrent state evolution. These methods improve tracking mainly by enriching spatial representations, introducing motion priors, or retaining more temporal context. However, these methods focus on representation enhancement, temporal-context aggregation, or motion modeling, without regulating correspondence reliability during recurrent state evolution.

Beyond enhancing representations or aggregating temporal context, SCRTrack regulates how target evidence is formed, propagated, and maintained throughout recursive tracking, thereby reducing the accumulation of unreliable spatial responses, ambiguous temporal associations, and distractor-induced state corruption. The MFF purifies shallow structural cues before selectively injecting them into deep semantic representations, rather than directly aggregating multiresolution features. The RCP regulates recurrent state evolution according to interframe correspondence reliability, rather than treating all propagated evidence equally. The MCP uses abnormal response-peak displacement to determine when appearance-based localization is unreliable and invokes motion-guided prediction only under potential drift.

B. Spatiotemporal Modeling for Visual Tracking

The spatiotemporal modeling is essential for maintaining tracking continuity under appearance variation, occlusion, and background interference. The spatiotemporal tracker [25] combined spatiotemporal context modeling with scale-adaptive

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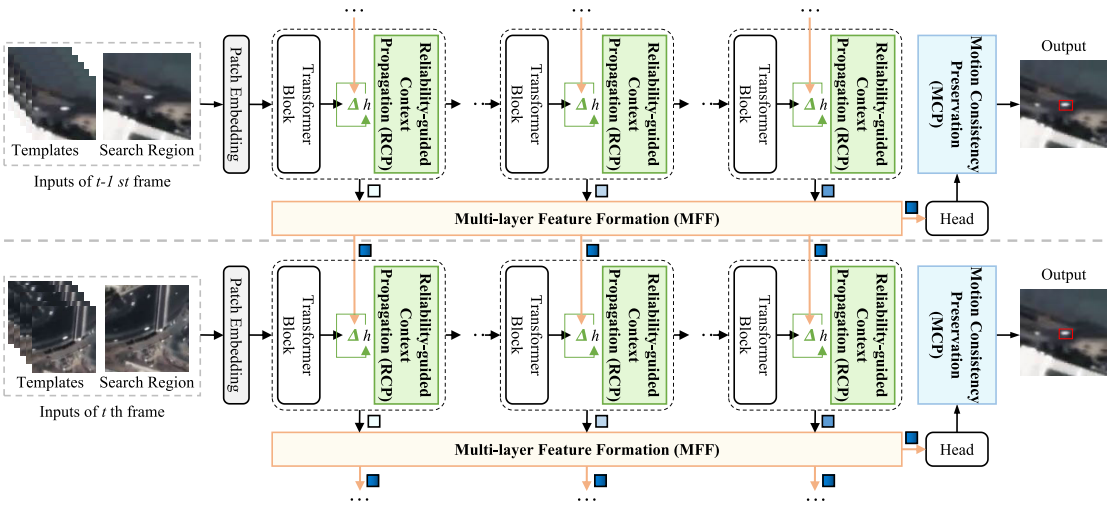


Fig. 2. Overview of the proposed SCRTrack for reliable feature propagation. First, the MFF module injects fine-grained cues into deep semantics to improve target discriminability. The blue blocks with different intensities distinguish features from different network depths, with lighter and darker blocks corresponding to shallower and deeper features, respectively. Then, the RCP module suppresses unreliable state updates by evaluating the reliability of interframe correspondences, where Δh denotes the reliability-regulated hidden-state increment during temporal state update. Finally, the MCP mechanism monitors target motion to ensure identity preservation under occlusion and background clutter.

correlation filtering, response-based occlusion detection, and extended Kalman filtering to improve tracking robustness under scale variation and occlusion. Recent trackers have moved beyond framewise appearance matching by incorporating historical information to enhance temporal consistency. The representative methods [4], [5] exploit attention mechanisms or autoregressive formulations to integrate target templates, search regions, and historical states. Other approaches [7], [26], [27], [28] further enhance temporal feature propagation, prompt-based historical guidance, or token-level memory construction. Despite their effectiveness, transformer-based trackers are often constrained by quadratic complexity when modeling video-level dependencies. The state-space models (SSMs), especially Mamba [29], have recently provided an efficient alternative for long-sequence modeling due to their linear complexity and recurrent hidden-state mechanism. In visual tracking, the Mamba-based methods [6], [30] exploit recurrent state propagation to capture long-term temporal context, achieving favorable efficiency and performance.

These methods demonstrate the importance of historical context for robust tracking. However, most existing spatiotemporal trackers focus on feature interaction or memory construction, while paying limited attention to the reliability of evidence being propagated through the tracking process. Different from the existing temporal fusion, historical memory, and recurrent state propagation, SCRTrack estimates tokenwise correspondence reliability by combining matching uncertainty with target confidence. The estimated reliability is mapped to the discretization step size of the state-space model, adaptively regulating how each token contributes to recurrent state evolution.

C. Conditional Update for Consistent Tracking

Consistent tracking requires not only discriminative target representation but also reliable regulation of when and how target states are updated. A common strategy is to update templates or target states using fixed intervals or confidence-based strategies. Early transformer and Siamese trackers adopt these strategies to update dynamic templates [16]. Although simple and effective, fixed updates are insensitive to abrupt appearance changes, while confidence-based updates may be misled by distractors with high response scores. This issue is particularly severe in satellite videos, where weak target appearance and cluttered backgrounds make ambiguous responses unreliable. Another line of work improves update robustness by maintaining or filtering historical information. The memory-based trackers store multiple historical target representations to provide long-term reference cues [14], [27]. Candidate elimination methods, such as OSTrack [31] and GRM [32], suppress less informative regions to reduce background interference. Recent temporal interaction methods further adapt templates or features by exploiting temporal coherence. For example, JTBP [33] adjusts template and search features with temporal consistency, while SiamTITP [13] refines template tokens with temporal information for more discriminative target representation. These methods reduce redundant or noisy information, but they regulate updates at a coarse level and rarely evaluate the reliability of cross-frame target correspondences.

Unlike coarse template or memory updates, SCRTrack regulates tokenwise state evolution according to correspondence reliability and preserves target identity when appearance-based localization becomes unreliable. Therefore, it aims to prevent unreliable evidence from being absorbed into the tracking state, rather than storing more historical cues or applying predefined update rules.

III. Proposed Method

A. Method Overview

To address spatiotemporal inconsistency accumulation in SVSOT, we propose SCRTrack, a spatiotemporal consistency regulation framework that regulates target evidence along the recursive tracking process. As shown in Fig. 2, SCRTrack consists of transformer blocks [34], a tracking head, and three regulation components: MFF, RCP, and MCP, which, respec-

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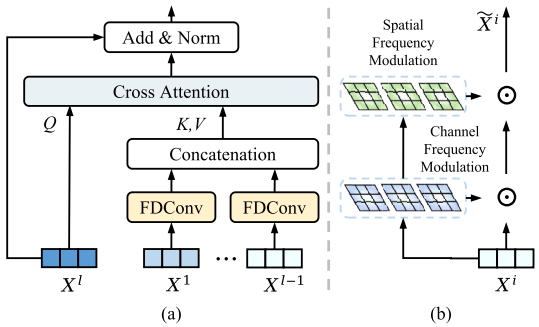


Fig. 3. Detailed architecture of the MFF module. (a) Multilayer fusion: deep semantic features X^l act as the query to selectively aggregate fine-grained details from multiple filtered shallow features via cross-attention. A learnable gating parameter γ adaptively regulates the injection of calibrated cues to suppress background clutter. (b) Frequency filter: each shallow feature is purified by an FDConv operator, which suppresses noise-sensitive and redundant frequency components before fusion.

tively, regulate spatial feature formation, temporal propagation, and identity-preserving state maintenance.

Given the template and search region, SCRTrack extracts multilevel token representations through transformer blocks. The MFF calibrates shallow structural cues and integrates them with deep semantic features to provide reliable spatial evidence for localization. The RCP propagates temporal information by estimating the correspondence reliability between current search tokens and previous regulated reference tokens, thereby suppressing unreliable recursive updates. The MCP further preserves motion consistency by adaptively balancing network predictions with motion-consistent estimates under occlusion or distractor interference. Together, these modules reduce unreliable evidence accumulation and maintain robust spatiotemporal consistency in degraded satellite videos.

B. MFF Module

To regulate spatial feature formation under weak observations, we propose an MFF module that integrates fine-grained structural cues into deep semantic features. Since shallow features retain structural details but are sensitive to clutter and noise, direct multilevel fusion may degrade semantic representations. The MFF addresses this by first purifying shallow features with frequency filters and then selectively injecting the refined cues into deep features through multilayer fusion, enhancing spatial discriminability while suppressing clutter interference. For clarity, we present the fusion strategy in Fig. 3(a) and the frequency-based purification in Fig. 3(b).

1) Multilayer Feature Calibration: We collect the outputs of l encoders, denoted as $[X^1, X^2, \ldots, X^l]$, where $X^l \in \mathbb{R}^{B\times N_x\times C}$, N_x is the number of search region tokens. To selectively inject fine-grained cues into deep semantic features, we take the deep feature X^l in the last layer as the query to extract low-level details from shallow features. We first purify each shallow feature X^i $(i = 1, \ldots, l-1)$ with a frequency filter and then concatenate them as $\tilde{X}_{\text{ml}} = \text{Concat}(\tilde{X}^1, \tilde{X}^2, \ldots, \tilde{X}^{l-1}) \in \mathbb{R}^{B\times((l-1)\times N_x)\times C}$. The module then performs multihead cross-attention with learnable projections

$$Q = \text{LN}\left(X^l\right) W_Q$$
$$K = \text{LN}\left(\tilde{X}_{\text{ml}}\right) W_K$$
$$V = \text{LN}\left(\tilde{X}_{\text{ml}}\right) W_V \tag{1}$$

where LN$(\cdot)$ denotes the layer normalization. The calibrated search features are computed as

$$X_{\text{ca}} = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{2}$$

where d_k is the scale parameter. We then fuse the original and calibrated features via a residual connection

$$X_{\text{MFF}} = \text{LN}\left(X^l + \gamma \odot X_{\text{ca}}\right) \tag{3}$$

where γ is a learnable channelwise gating vector that regulates the contribution of the multilayer details relative to the deep semantics. In the SCRTrack configuration, γ is initialized to zero and optimized during training. Layer normalization is applied after fusion to stabilize optimization. In this way, MFF regulates multilayer injection of fine-grained cues to suppress clutter regions, improving localization robustness in satellite scenes.

2) Frequency Filter for Detail Enhancement: We adopt FDConv [35] as a frequency-aware filtering operator to purify shallow features before multilayer fusion. For each shallow feature $X^i \in \mathbb{R}^{B\times N_x\times C}$, we reshape it to a 2-D feature map $U^i \in \mathbb{R}^{B\times C\times H\times W}$, where $N_x = H \times W$, and apply FDConv to obtain the filtered feature map.

FDConv constructs frequency-diverse kernels through disjoint Fourier weights and modulates them from two complementary perspectives. The channel frequency modulation uses local and global channel contexts to adjust individual kernel elements, whereas spatial frequency modulation reweights different frequency bands according to local feature content. Their combination does not uniformly suppress high-frequency components, which may contain both target boundaries and imaging noise, but adaptively retains target-related structural cues while attenuating spatially irrelevant and channel-redundant responses caused by background clutter and low-quality imaging.

Finally, $\tilde{U}^i = \text{FDConv}(U^i)$ is reshaped back to token form to obtain $\tilde{X}^i$, which is used to build $\tilde{X}_{\text{ml}}$ in (1).

Through frequency-aware purification and gated multilayer fusion, the MFF regulates the formation of spatial features before it enters the recursive tracking process. This design enhances reliable fine-grained target structures while reducing clutter interferences, providing a trustworthy spatial basis for temporal context regulation.

C. RCP Module

To regulate temporal propagation under unreliable interframe associations, we propose RCP. The RCP estimates the correspondence distribution between current search tokens and temporally aggregated reference tokens, using it both as an alignment prior for short-term spatial anchoring and to derive a reliability score for modulating Mamba temporal state evolution. In this way, historical context is propagated only when cross-frame correspondence is reliable. The structure of RCP is shown in Fig. 4.

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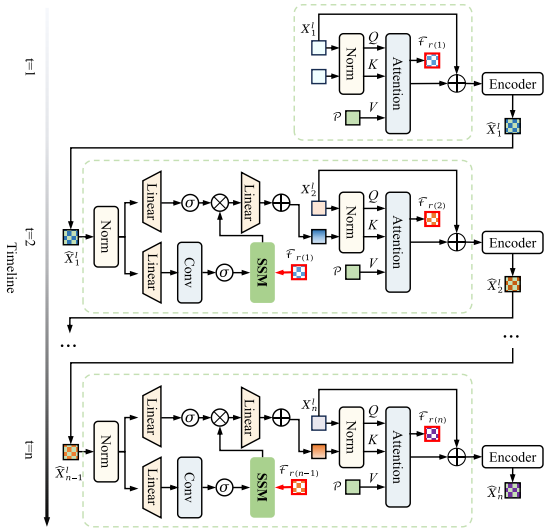


Fig. 4. Illustration of the RCP module. RCP estimates the correspondence distribution between current search tokens and temporally aggregated reference tokens. The distribution provides an alignment prior for short-term anchoring and yields correspondence reliability to modulate the Mamba discretization step size for reliability-aware temporal propagation. At $t = 1$, the temporal state is initialized with the current-frame tokens.

1) Temporal Correspondence Distribution: Let $X_t^l \in \mathbb{R}^{B\times N_x\times C}$ denote the search tokens from the lth encoder layer at time t, and let $X_{t-1}^{\mathrm{AT}} \in \mathbb{R}^{B\times N_x\times C}$ denote the temporally aggregated reference tokens from the previous time step. We compute the interframe correspondence distribution by scaled dot-product matching

$$Q_t = \mathrm{LN}\left(X_t^l\right) W_Q, K_{t-1} = \mathrm{LN}\left(X_{t-1}^{\mathrm{AT}}\right) W_K \tag{4}$$

$$\mathcal{F}_{t-1\to t} = \mathrm{Softmax}\left(\frac{Q_t K_{t-1}^{\top}}{\sqrt{d_k}}\right). \tag{5}$$

We then introduce a coordinate map $\mathcal{P}(x, y) \in [-1, 1]^2$, which is embedded and reshaped into $\mathcal{P} \in \mathbb{R}^{N_x\times C}$. Based on the correspondence distribution, we compute the alignment prior

$$\mathcal{P}_{t-1\to t} = \mathcal{F}_{t-1\to t}\mathcal{P}. \tag{6}$$

The alignment prior is further injected into the search region tokens via a residual connection

$$\bar{X}_t^l = X_t^l + \mathcal{P}_{t-1\to t}. \tag{7}$$

$\bar{X}_t^l$ is then fed into the next encoder layer for further processing, providing a stable short-term position for subsequent temporal modeling.

2) Adaptive Temporal Scale Mamba: To derive the temporal context of reliable regions, we leverage the correspondence distribution to calculate the reliability of different regions and regulate the discretization step size of the SSM based on the reliability. We first obtain the correspondence uncertainty by calculating the entropy of the correspondence distribution

$$\tilde{H}_{\mathcal{F}}(n) = \frac{-\sum_{m=1}^{N_x} \mathcal{F}_{t-1\to t}^{(n,m)} \log\left(\mathcal{F}_{t-1\to t}^{(n,m)} + \epsilon\right)}{\log N_x} \tag{8}$$

where ϵ is a small constant for numerical stability, and $\tilde{H}_{\mathcal{F}}(n)$ is computed for each query token n, yielding a tokenwise uncertainty estimate.

Since correspondence concentration alone may be unreliable under occlusion or distractor interference, we further introduce the target confidence $H_c^{\max}$, obtained as the maximum value of the target center score map. The tokenwise correspondence reliability is defined as

$$\mathcal{F}_r(n) = \left(1 - \tilde{H}_{\mathcal{F}}(n)\right) H_c^{\max}. \tag{9}$$

The reliability metric is motivated by the ambiguity of discrete correspondence estimation. We use normalized Shannon entropy because it measures the dispersion of the correspondence distribution. When the matching evidence is concentrated on one reference location, $\tilde{H}_{\mathcal{F}}(n)$ approaches zero, indicating an unambiguous match. As the evidence becomes distributed over multiple candidate locations, the entropy increases and reaches one for a uniform distribution. Therefore, $1 - \tilde{H}_{\mathcal{F}}(n)$ serves as a tokenwise measure of correspondence concentration.

However, a concentrated distribution only indicates low matching ambiguity and does not guarantee that the selected correspondence belongs to the target. When occlusion makes the correspondence distribution diffuse or weakens the target center response, the resulting reliability score decreases. This score is subsequently used to condition the SSM discretization step size through a learned mapping. We therefore combine correspondence concentration with the frame-level localization confidence $H_c^{\max}$. The two terms characterize complementary conditions for reliable propagation: the former measures whether a query token has an unambiguous temporal match, whereas the latter measures whether the current frame contains a confident target localization. Their product acts as a soft conjunction, assigning high reliability only when both conditions are satisfied.

We then use $\mathcal{F}_r(n)$ to regulate the state evolution in Mamba. Following the standard Mamba [29] design, the SSM models temporal information via a hidden state h_t that evolves

$$\begin{aligned} h_t &= \bar{\mathbf{A}} h_{t-1} + \bar{\mathbf{B}} x_t \\ y_t &= \mathbf{C} h_t \end{aligned} \tag{10}$$

where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ are the discretized parameters obtained by matrix exponential discretization with step size $\Delta_{\mathcal{F}}$

$$\begin{aligned} \bar{\mathbf{A}} &= \exp\left(\Delta_{\mathcal{F}}\mathbf{A}\right) \\ \bar{\mathbf{B}} &= \left(\Delta_{\mathcal{F}}\mathbf{A}\right)^{-1}\left(\exp\left(\Delta_{\mathcal{F}}\mathbf{A}\right) - \mathbf{I}\right) \cdot \Delta_{\mathcal{F}}\mathbf{B} \approx \Delta_{\mathcal{F}}\mathbf{B}. \end{aligned} \tag{11}$$

The discretization step size is regulated tokenwise. Specifically, for each query token n, we predict an individual step size $\Delta_{\mathcal{F}}(n)$ and use it to modulate the corresponding state update

$$\Delta_{\mathcal{F}}(n) = \mathrm{Softplus}\left(\Delta + \mathrm{Linear}\left(\mathcal{F}_r(n)\right)\right) \tag{12}$$

where Δ is the learnable base step size in Mamba, and Linear$(\cdot)$ maps the reliability to a bias. In this way, reliable correspondences are allowed to contribute to temporal state propagation, while ambiguous associations are adaptively weakened by the regulated SSM update.

Overall, the RCP couples correspondence-aware alignment and reliability-guided state evolution within the same temporal regulation process. It anchors short-term localization and suppresses the accumulation of mismatched historical evidence, providing reliable temporal propagation for robust SVSOT.

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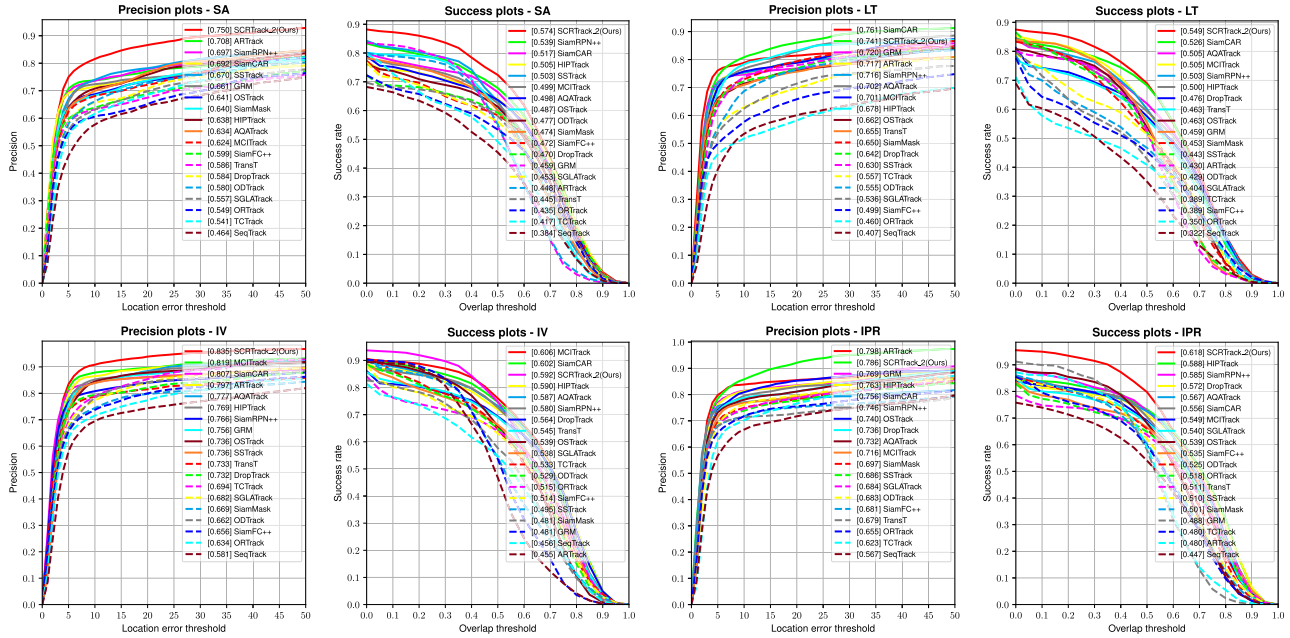


Fig. 5. Attribute-based precision and success rates of trackers on OOTB.

TABLE I

PERFORMANCE COMPARISON OF SV TRACKERS ON SATSOT AND SV248S. THE BEST RESULTS ARE HIGHLIGHTED IN RED, THE SECOND-PLACE RESULTS ARE HIGHLIGHTED IN BLUE, AND THE THIRD-PLACE IN GREEN. GV INDICATES GENERAL VIDEO, UV INDICATES UAV VIDEO, AND SV INDICATES SATELLITE VIDEO

Trackers	Source	Architecture	Training Dataset	SatSOT		SV248S		OOTB		FPS
				P.(%)	S.(%)	P.(%)	S.(%)	P.(%)	S.(%)	
CFME [17]	TGRS2020	CF-based	-	55.7	41.0	68.8	42.7	75.9	61.1	270.7
HRSiam [15]	TIP2021	CNN-based	GV	53.4	42.1	52.6	29.9	73.0	55.3	84.5
SiamMDM [16]	TGRS2023	CNN-based	GV	61.1	46.1	72.5	45.6	-	-	61.5
TATrans [36]	TGRS2024	ViT-based	GV	57.6	45.6	-	-	66.4	51.8	-
MACF [18]	TGRS2024	CF-based	-	65.3	49.4	-	-	-	-	11.7
REPS [37]	JAG2024	CNN-based	GV	-	-	-	-	68.9	45.4	11.2
SiamS^{2}F [19]	TGRS2025	CNN-based	GV	62.1	48.6	76.3	50.6	80.7	48.6	82.2
HLSOT [23]	JSTARS2025	CNN-based	GV	66.9	53.1	77.0	53.7	84.0	65.2	70.1
SiamTITP [13]	TIP2025	CNN-based	GV	66.5	51.3	83.4	54.4	82.7	63.7	33.9
MTTrack [24]	KBS2026	Mamba&ViT-based	GV	68.7	47.9	79.2	58.4	-	-	101.9
SCRTrack#2	Ours	Mamba&ViT-based	GV	69.2	53.6	75.7	49.2	82.6	61.3	66.1
SVLPNet [38]	TCSVT2024	ViT-based	SV	58.4	46.5	76.9	47.1	-	-	-
EHTracker [39]	JSTARS2025	ViT-based	SV	61.6	48.2	74.3	49.0	-	-	-
STAR [12]	TGRS2025	ViT-based	GV+SV	63.9	53.7	77.4	52.3	84.6	67.8	87.3
MemTrack [14]	ISPRS2025	ViT-based	GV+UV	74.0	57.0	83.7	54.3	88.7	71.2	43.8
TAGET [40]	TGRS2026	CNN-based	SV	65.6	49.5	77.1	48.6	-	-	48.6
TSTrans [41]	TGRS2026	ViT-based	SV	61.4	53.2	80.6	40.8	-	-	49.2
SCRTrack	Ours	Mamba&ViT-based	GV+SV	76.3	60.0	83.2	57.1	92.5	70.7	66.1

D. MCP Mechanism

To maintain target identity under occlusion and similar-object interference, we introduce MCP, which monitors response-peak displacement across motion scales. In satellite videos, target motion may vary globally, but peak displacement is usually locally consistent within a short temporal window, and abrupt deviations indicate potential identity drift. Unlike motion-based recovery [17], the MCP uses motion consistency as a reliability cue. It preserves the appearance-based prediction when the peak follows recent motion statistics and switches to motion-guided prediction when sudden shifts occur, thereby improving identity continuity.

The predicted target center at frame t is obtained by locating the maximum response P_t in the score map H_c. The score map H_c is generated by the tracking head, which takes the output of the MFF module as an input. To quantify localization stability, we compute the frame-to-frame displacement of response peaks as

$$D_t = \|P_t - P_{t-1}\|_2. \tag{13}$$

To adapt to different motion scales, we further estimate an adaptive drift threshold τ_t by averaging the peak displacements

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over the most recent I frames

$$\tau_t = \frac{1}{I}\sum_{k=t-I}^{t-1} D_k. \tag{14}$$

For the first I frames, we compute τ_t using all reliable past displacements. τ_t characterizes the peak relocation amplitude when the tracker remains locked to the same target instance, and it is violated when occlusion causes abrupt peak switching.

At each frame t, the tracker maintains two candidate target states: 1) an appearance-based neural prediction O_t^{NN} regressed from deep semantic features and 2) a motion-based prediction O_t^{KF} generated by the KF. When appearance-based localization becomes unreliable, the MCP dynamically adjusts the preference between appearance-driven and motion-based predictions, enabling stable target localization. The MCP switches to the KF branch when the peak relocation exceeds the adaptive baseline

$$\bar{O}_t = \begin{cases} O_t^{\mathrm{KF}}, & \text{if } D_t > \tau_t \\ O_t^{\mathrm{NN}}, & \text{otherwise.} \end{cases} \tag{15}$$

To avoid frequent switching caused by noisy responses, we further require the tracker to remain stable for I consecutive frames ($D_t \le \tau_t$) before switching back to neural predictions. In this manner, the MCP preserves neural predictions under normal conditions while preventing abrupt peak switching from propagating into long-term identity drift. Based on the study in Section IV-D, we set $I = 10$ in all experiments, which provides an appropriate tradeoff between suppressing response noise and preserving adaptability to local motion variations.

As a result, the MCP preserves appearance-based localization under normal conditions while providing motion-consistent identity maintenance under occlusion and distractor interference. Together with MFF and RCP, the MCP completes the reliability-aware regulation of SCRTrack that MFF calibrates spatial evidence formation, RCP regulates temporal evidence propagation, and MCP maintains identity-consistent target states when appearance evidence becomes unreliable.

IV. Experiments

A. Experimental Setup

This section documents the experimental setup to evaluate SCRTrack under weak observations and significant appearance variations. We utilize three widely used satellite video datasets, SatSOT [42], SV248S [43], and OOTB [44], to evaluate the performance of our proposed tracker.

1) Implementation Details: The model is trained in two stages. First, it is trained for 100 epochs on four large-scale general video tracking datasets, i.e., GOT-10k [45], TrackingNet [46], LaSOT [47], and COCO [48]. AdamW [49] is used with learning rates of 4×10^{-5} for the backbone and 4×10^{-4} for other parameters, and a weight decay of 1×10^{-4} activated after the 80th epoch. Following STAR [12], the model is then fine-tuned for 50 epochs on SatSOT-train [22], with weight decay activated after the 30th epoch. SatSOT-train is derived from SatMTB [22] and contains 947 sequences from four categories. The search region and template sizes are set to 224×224 and 112×112, respectively. Training is conducted on four NVIDIA RTX 4090 GPUs with a total batch size of 64. All tests are conducted on a single NVIDIA RTX 4090 GPU.

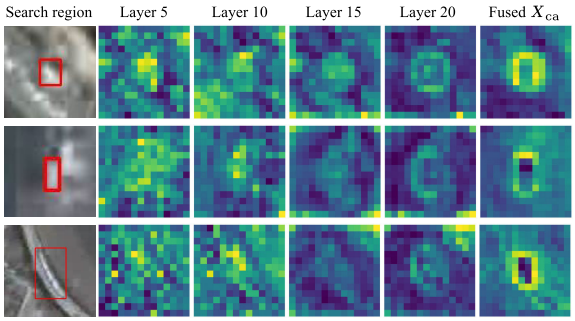


Fig. 6. Visualization of attention maps in different layers. The shallow layers capture more target details, whereas the deep layers better extract target semantic discrimination. The fused shallow and deep features yield clear feature representations.

2) Evaluation Metrics: We follow the standard single object tracking (SOT) evaluation protocol [50] using one-pass evaluation (OPE) with precision and success metrics. Considering the small target sizes in satellite videos, precision is measured as the percentage of frames whose predicted center is within 5 pixels of the ground-truth center, following [42]. Success is evaluated by the AUC of the success plot, which reports the proportion of frames whose IoU exceeds thresholds ranging from 0 to 1.

B. Comparison With the State-of-the-Arts Methods

1) Comparison With Satellite Video Trackers: We compare SCRTrack with representative satellite video trackers on SatSOT, SV248S, and OOTB, as shown in Table I. To account for different training data, we report SCRTrack trained with both GV and SV data and SCRTrack#2 trained only with GV data. SCRTrack achieves consistent gains across all benchmarks. On SatSOT, it obtains the highest rates of 76.3% precision and 60.0% success. On SV248S, it reaches a success rate of 57.1%. On OOTB, it ranks first in precision with 92.5% and second in success with 70.7%. Moreover, SCRTrack#2 outperforms HRSiam and MCITrack by 15.8%/11.5% and 18.5%/13.8% in precision/success on SatSOT, respectively. Compared with the recent Mamba-based MTTrack, SCRTrack#2 improves precision/success by 0.5%/5.7% on SatSOT, demonstrating that the gains come not only from satellite-domain training data but also from the proposed spatiotemporal consistency regulation.

2) Comparison With General Video Trackers: We further compare SCRTrack with representative general video trackers. As shown in Table II, SCRTrack achieves the best results on all three benchmarks in terms of both precision and success rate. On SatSOT, SCRTrack improves the performance to 69.2% precision and 53.6% success, respectively. On SV248S, where low resolution and background clutter cause severe performance degradation for most general trackers, SCRTrack achieves 75.7% precision and 49.2% success. On OOTB, SCRTrack obtains 82.6% precision and 61.3% success, achieving the best tracking performance. Compared with the foundation-model-based trackers, SCRTrack#2 outperforms SAM2 and SAMURAI on all three satellite benchmarks. These

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TABLE II

PERFORMANCE COMPARISON OF GV TRACKERS ACROSS DIFFERENT DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN RED, THE SECOND-PLACE RESULTS ARE HIGHLIGHTED IN BLUE, AND THE THIRD-PLACE IN GREEN

Trackers	Source	SatSOT		SV248S		OOTB		FPS
		P.(%)	S.(%)	P.(%)	S.(%)	P.(%)	S.(%)	
SiamRPN++ [8]	CVPR2019	56.0	40.9	74.3	46.3	77.2	59.4	85.9
SiamCAR [9]	CVPR2020	59.8	43.9	70.4	43.3	81.9	60.9	66.3
SiamFC++ [10]	AAAI2020	50.6	37.6	61.7	37.6	68.2	53.7	245.6
TCTrack [26]	CVPR2022	46.3	32.2	34.0	13.5	65.7	50.2	122.5
OSTrack [31]	ECCV2022	53.9	37.7	52.6	20.9	74.7	54.8	100.6
SeqTrack [4]	CVPR2023	45.8	35.4	21.9	9.4	60.4	47.0	6.1
ARTrack [5]	CVPR2023	52.4	33.8	49.9	18.6	80.7	47.5	6.5
ODTrack [7]	AAAI2024	56.4	42.6	45.1	18.2	69.1	54.3	40.1
HIPTrack [27]	CVPR2024	54.4	42.0	33.4	15.5	78.0	60.1	50.9
SAMURAI [51]	Arxiv2024	43.1	36.0	39.6	23.8	59.1	37.4	25.6
MCITrack [6]	AAAI2025	50.7	39.8	55.9	22.1	78.5	59.2	99.1
ORTrack [52]	CVPR2025	48.0	38.0	39.4	17.5	65.1	52.1	33.3
SGLATrack [53]	CVPR2025	50.3	38.8	41.7	18.6	68.4	53.9	142.9
SAM2 [54]	ICLR2025	51.2	42.0	45.9	27.6	67.8	42.7	40.6
SCRTrack#2	Ours	69.2	53.6	75.7	49.2	82.6	61.3	66.1

TABLE III

ATTRIBUTE-BASED SUCCESS RATES OF TRACKERS ON SATSOT. THE BEST RESULTS ARE HIGHLIGHTED IN RED, THE SECOND-PLACE RESULTS ARE HIGHLIGHTED IN BLUE, AND THE THIRD-PLACE IN GREEN

Trackers	ARC	BC	BJT	DEF	FOC	IV	LQ	POC	ROT	SOB	TO
GRM	32.4	17.7	27.6	41.6	18.3	35.2	20.2	22.9	24.8	12.6	5.1
TCTrack	29.1	20.6	32.0	26.4	22.0	45.7	25.2	26.2	27.5	19.9	12.0
ARTrack	33.1	25.2	30.8	32.7	25.6	38.6	30.5	25.3	28.7	26.8	14.5
TransT	29.1	20.2	31.9	34.8	32.3	39.0	22.5	28.4	29.5	22.5	12.4
SeqTrack	38.4	23.2	32.7	37.3	26.1	34.2	23.7	28.3	30.9	19.6	9.5
SiamMask	28.1	25.4	34.9	31.3	23.0	51.3	35.3	30.7	32.8	24.9	17.4
SSTrack	26.4	23.6	37.2	24.1	21.7	38.5	33.4	29.4	34.4	28.8	16.4
SiamFC++	30.3	25.5	35.6	31.0	23.8	42.3	31.7	31.5	35.0	27.0	16.5
OSTrack	31.4	30.0	34.3	32.9	28.9	40.2	36.9	26.7	36.7	29.5	18.1
ORTrack	36.3	25.2	33.7	41.5	31.8	44.0	28.3	27.1	33.4	21.0	8.8
SGLATrack	36.9	25.0	43.7	40.2	33.1	50.4	34.2	28.0	31.6	24.3	11.1
MCITrack	52.9	30.2	36.2	51.0	30.3	57.8	25.4	30.3	34.2	18.8	9.2
DropTrack	46.6	28.5	38.2	43.3	35.0	48.0	31.2	33.4	34.4	22.4	10.8
SiamRPN++	33.8	31.2	47.5	43.4	29.2	56.3	44.2	29.9	37.5	34.4	18.9
AQATrack	45.4	28.9	39.6	42.9	33.8	53.8	31.1	31.6	35.0	24.3	14.9
HIPTrack	50.2	31.4	38.8	49.4	33.4	50.1	32.7	34.0	35.7	23.9	10.6
ODTrack	49.5	33.0	40.6	46.1	31.3	41.3	33.5	30.2	38.6	25.5	9.8
SiamCAR	27.4	32.8	51.8	35.4	32.7	57.9	52.3	31.9	41.0	39.5	33.7
SCRTrack#2	54.4	44.3	52.5	49.1	49.4	72.8	53.3	46.2	48.9	49.0	38.8

results show that directly applying the evaluated pretrained SAM2 and SAMURAI configurations remains challenging on these satellite benchmarks. These results indicate that general video trackers are less robust to satellite-specific degradation, especially when target appearance is weak and interframe associations are unreliable. By regulating spatial evidence formation, temporal evidence propagation, and identity-preserving state maintenance, SCRTrack provides stable tracking under challenging SVSOT scenarios.

C. Attribute-Based Comparison

To further analyze the robustness of our method under satellite challenges, we conduct attribute-based comparison on SatSOT, SV248S, and OOTB. For fair comparison with general video trackers, all results in Tables III and IV and Fig. 5 are reported using SCRTrack#2, which is trained only on GV data.

1) Tracking Under Unreliable Appearance: Appearance evidence becomes unreliable when targets are tiny or surrounded by cluttered backgrounds. Under these conditions, target representations extracted from satellite videos are easily contaminated by background responses, making spatial evidence unstable. As shown in Table III, SCRTrack#2 achieves the best success rates on all appearance-related attributes in SatSOT, including low quality (LQ, 53.3%), tiny object (TO, 38.8%), and background clutter (BC, 44.3%). Similar trends are observed on SV248S in Table IV. SCRTrack#2 achieves the best score on dense similarity (DS, 49.5%) and also ranks second on background cluster (BCL, 38.5%). On OOTB in Fig. 5, SCRTrack#2 achieves strong results on appearance-related attributes, including less texture (LT,

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TABLE IV

ATTRIBUTE-BASED SUCCESS RATES OF TRACKERS ON SV248S. THE BEST RESULTS ARE HIGHLIGHTED IN RED, THE SECOND-PLACE RESULTS ARE HIGHLIGHTED IN BLUE, AND THE THIRD-PLACE IN GREEN

Trackers	BCH	BCL	CO	DS	IPR	IV	LTO	ND	SM	STO
SeqTrack	10.7	4.5	5.6	7.3	6.7	6.9	12.3	4.9	8.2	5.7
TCTrack	17.1	6.6	8.2	11.4	9.4	9.5	13.2	5.6	10.6	10.0
DropTrack	18.1	8.3	8.8	12.8	10.7	9.8	12.7	11.2	11.2	10.7
HIPTrack	18.4	8.9	8.4	13.3	11.3	9.8	14.8	12.1	11.5	10.3
ORTrack	20.5	8.2	9.7	15.1	12.7	12.9	14.1	10.9	11.3	13.5
ODTrack	20.7	10.9	10.2	16.5	13.7	13.2	15.5	15.6	13.9	12.5
ARTrack	20.0	12.4	10.6	17.3	15.6	13.9	12.9	13.7	16.9	13.7
SGLATrack	21.3	9.3	10.0	16.7	13.9	12.1	17.0	14.9	13.7	13.5
SSTrack	22.3	12.5	10.8	16.8	14.5	13.7	16.8	12.1	13.0	14.8
GRM	21.3	9.6	9.5	17.6	14.3	14.6	14.1	10.7	14.3	14.7
AQATrack	22.4	12.3	10.0	17.2	15.6	14.1	16.2	16.1	16.4	13.7
OSTrack	24.0	13.1	9.7	19.7	16.5	14.8	14.4	15.1	17.1	15.1
MCITrack	23.9	13.9	13.5	19.9	17.1	16.3	16.8	14.1	14.6	16.6
SiamMask	24.8	16.6	12.2	22.3	20.0	17.8	13.5	16.7	21.1	17.4
TransT	30.5	18.1	18.6	26.2	23.7	23.2	20.8	11.7	24.7	22.7
SiamFC++	36.5	32.0	22.3	36.5	33.3	35	31.1	27.9	34.2	24.6
SiamCAR	44.8	36.5	24.6	42.5	39.1	38.5	32.2	35.9	42.9	29.8
SiamRPN++	47.4	39.4	28.2	46.1	42.3	41.1	33.8	34.5	45.4	32.8
SCRTrack#2	48.5	38.5	27.7	49.5	44.4	42.3	39.1	41.2	47.8	33.7

54.9%) and illumination variation (IV, 59.2%), ranking among the top methods. These results indicate that MFF improves spatial evidence formation by calibrating shallow cues and suppressing background-induced noise.

2) Tracking Under Ambiguous Correspondence: Ambiguous correspondence arises when targets undergo ROT, aspect-ratio variation, background jitter, or subtle interframe motion. In these cases, cross-frame matching becomes uncertain, and blindly propagating historical features may accumulate mismatched temporal evidence. Table III shows that SCRTrack#2 achieves the best performance on rotation (ROT, 48.9%), aspect ratio change (ARC, 54.4%), and background jitter (BJT, 52.5%), indicating robustness to viewpoint and geometry variations. On SV248S in Table IV, SCRTrack#2 ranks first on natural disturbance (ND, 41.2%), in-plane rotation (IPR, 44.4%), and background change (BCH, 48.5%). Correspondence-conditioned temporal regulation prevents the blind propagation of mismatched historical states and helps maintain stable trajectories under subtle motion patterns. On OOTB, which contains diverse motion patterns and annotation styles, SCRTrack#2 remains competitive on related attributes. In particular, it ranks first on in-plane rotation (IPR, 61.8%), as shown in Fig. 5. The gains suggest that RCP effectively regulates temporal propagation by estimating correspondence reliability and suppressing mismatched historical evidence.

3) Tracking Under Occlusion: Occlusion is particularly challenging in satellite video tracking because targets may become partially or fully invisible, while nearby distractors with similar appearance can dominate the response map. In such cases, the appearance-based localization is prone to abrupt peak switching, which may further cause long-term identity drift. As shown in Table III, SCRTrack#2 achieves the best success rates under partial occlusion (POC, 46.2%) and full occlusion (FOC, 49.4%), showing strong recovery capability after severe occlusions. On SV248S in Table IV, SCRTrack#2 ranks first under occlusion attributes, including long-term occlusion (LTO, 39.1%) and short-term occlusion (STO, 33.7%), and ranks second under continuous occlusion (CO, 27.7%). These results demonstrate that MCP improves identity preservation by detecting inconsistent response-peak displacement and switching to motion-guided prediction when appearance localization becomes unreliable.

Overall, the attribute-based analysis confirms that MFF, RCP, and MCP address complementary sources of unreliability in satellite video tracking. The MFF improves spatial evidence formation under degraded appearance, RCP stabilizes temporal evidence propagation under ambiguous correspondence, and MCP preserves identity-consistent target states under occlusion. Together, these components form a reliability-aware spatiotemporal regulation framework for robust recursive satellite video tracking.

D. Ablation Study

We conduct comprehensive ablation studies to analyze how each component of SCRTrack contributes to robust satellite video single-object tracking under unreliable observations. The baseline is a one-stream tracker using Fast-iTPN [34] as the backbone.

1) Spatial Reliability Under Weak Observations: The MFF aims to enhance spatial evidence formation under ambiguous target appearances. As shown in Table V, the MFF improves appearance-related attributes, increasing the success rate from 62.3% to 62.9% on low quality (LQ) and from 51.9% to 54.0% on dense similarity (DS). This indicates that calibrated multilayer fusion helps recover discriminative details in low-resolution and cluttered scenes.

We further study the selected layers for fusion, using the last layer (Layer 20) as the query feature. As reported in Table VI, the three-layer configuration achieves the best success rate. The partial-layer combinations bring limited gains, as insufficient shallow cues may lose localization details, while uncalibrated low-level responses may introduce background

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TABLE V
ABLATION STUDIES OF DIFFERENT VARIANTS ON SATSOT AND SV248S WITH RELEVANT ATTRIBUTES

Method	MFF	RCP	MCP	SatSOT		SV248S		LQ	DS	ROT	ND	LTO	FOC
				P.(%)	S.(%)	P.(%)	S.(%)	S.(%)	S.(%)	S.(%)	S.(%)	S.(%)	S.(%)
Baseline				71.2	56.0	75.9	52.6	62.3	51.9	53.5	47.2	33.6	34.3
+ Spat. Reg.	✓			73.8	58.2	77.8	54.2	62.9	54.0	55.3	51.5	33.6	34.4
+ Temp. Reg.		✓		74.3	58.0	77.7	54.8	62.1	52.7	55.7	53.5	32.5	34.7
+ Spat-Temp. Reg.	✓	✓		75.6	58.9	81.0	56.0	62.8	55.8	56.4	48.8	34.3	34.5
+ Prediction			✓	74.6	57.6	79.2	54.4	62.8	53.6	54.7	48.0	39.9	42.4
SCRTrack	✓	✓	✓	76.3	60.0	83.2	57.1	63.4	57.4	57.4	56.1	38.9	43.3

TABLE VI
ABLATION STUDIES ON LAYER INDEX COMBINATIONS

Layer Index	SatSOT		SV248S	
	P.(%)	S.(%)	P.(%)	S.(%)
[5]	72.2	56.6	81.0	55.6
[10]	74.1	57.8	78.6	53.6
[15]	75.2	57.8	81.1	55.6
[5, 10]	74.1	57.7	78.9	54.3
[10, 15]	74.5	56.1	79.7	53.2
[5, 10, 15] (SCRTrack)	76.3	60.0	83.2	57.1

TABLE VII
ABLATION STUDIES BETWEEN OURS AND SIMPLE FEATURE FUSION STRATEGIES

Method	SatSOT		SV248S	
	P.(%)	S.(%)	P.(%)	S.(%)
w/ Add	73.9	56.1	80.1	55.0
w/ Average	74.6	57.7	80.2	55.5
w/o FF	72.5	57.6	82.0	56.6
fixed $\gamma = 1$	75.0	58.2	81.3	56.7
SCRTrack	76.3	60.0	83.2	57.1

TABLE VIII
ANALYSIS OF THE INITIALIZATION OF THE LEARNABLE GATE γ ON SATSOT

γ Initialization	0	0.01	0.1	0.5	1
P.(%)	76.3	75.2	75.7	74.9	75.2
S.(%)	60.0	58.4	58.6	58.1	58.1

noise. Thus, the MFF benefits from complementary multilayer information to form more reliable spatial representations.

Table VII compares different fusion strategies. On SatSOT, addition and average fusion achieve 73.9%/56.1% and 74.6%/57.7% in precision/success, respectively, but their fixed rules cannot effectively separate target details from clutter. Removing the frequency filter also degrades performance, demonstrating the importance of frequency-aware purification for suppressing noisy shallow features.

Compared with the nonlearnable setting of $\gamma = 1$, the learnable gate improves the precision/success rates by 1.3%/1.8% and 1.9%/0.4% on SatSOT and SV248S, respectively, demonstrating the benefit of adaptively learning channelwise residual scaling rather than using a fixed scale. Note that the fixed $\gamma = 1$ in Table VII is kept constant throughout training, whereas the settings in Table VIII use a learnable γ with different initial values. We further evaluate learnable γ initialized to 0, 0.01, 0.1, 0.5, and 1.0 in Table VIII. Zero initialization achieves the best performance, improving precision by at least 0.6% and success by at least 1.4% over the nonzero initializations. With $\gamma = 0$, MFF initially preserves the deep representation and progressively learns the contribution of the cross-attention residual, avoiding premature injection of clutter-sensitive shallow cues. Therefore, we adopt a learnable γ initialized to zero in the final SCRTrack configuration.

Fig. 6 further shows that the calibrated feature X_{ca} preserves target-related fine details while suppressing background responses, validating the effectiveness of MFF in complex and low-resolution satellite scenarios.

2) Temporal Reliability Under Ambiguous Correspondences: The RCP mitigates unreliable temporal propagation caused by ROT, deformation, and background motion. As shown in Table V, enabling RCP improves success by 2.2% on rotation (ROT) and 6.3% on natural disturbance (ND), demonstrating its effectiveness in regulating state updates under temporal appearance variations.

TABLE IX
ABLATION STUDIES OF DIFFERENT TEMPORAL CONTEXT REGULATION

Method	SatSOT		SV248S	
	P.(%)	S.(%)	P.(%)	S.(%)
w/o $\bar{\mathcal{P}}_{t-1\to t}$	74.1	57.9	79.6	54.8
w/o $\mathcal{F}_r$	75.4	58.4	80.0	54.9
SCRTrack	76.3	60.0	83.2	57.1

Table IX further validates the design of RCP. Removing the correspondence reliability $\mathcal{F}_r$ leads to success drops of 1.6% and 2.2% on SatSOT and SV248S, respectively, while removing the alignment prior also degrades performance. These results indicate that naive temporal aggregation is insufficient for satellite tracking, and both correspondence reliability and alignment prior are necessary for stable temporal state evolution.

Fig. 7 illustrates the behavior of RCP under challenging variations. When abrupt aspect-ratio changes or ambiguous correspondences occur, the RCP suppresses unreliable updates to prevent temporal memory corruption. Once the target becomes distinguishable again, the preserved state enables

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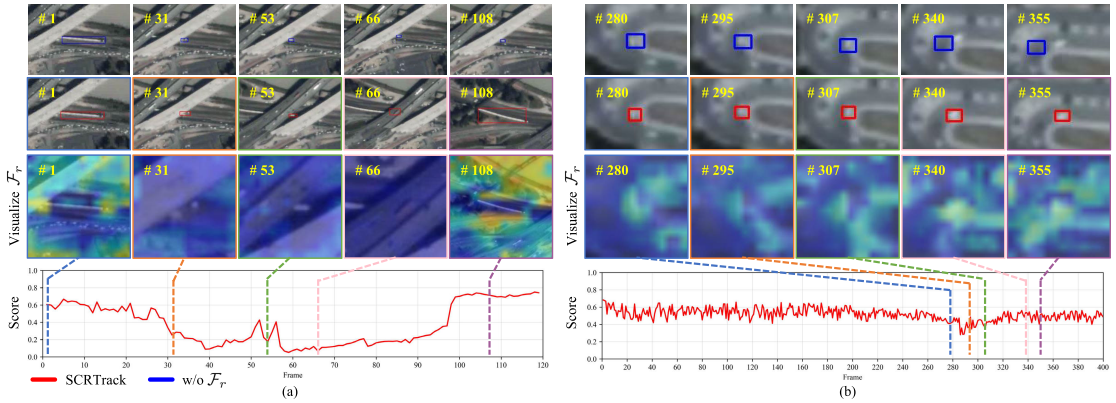


Fig. 7. Tracking results under complex scenarios with visualized interframe correspondence reliability map $\mathcal{F}_r$. (a) Shows a sequence with multiple occlusions, where the train is partially or fully occluded by bridges. (b) Shows a sequence with severe in-plane rotations, where the target appearance changes drastically and becomes ambiguous. Our RCP module adaptively modulates the temporal update strength based on the reliability of interframe correspondences, thereby suppressing unreliable updates under occlusion or ambiguous appearance and maintaining a stable tracking state.

TABLE X

ABLATION STUDIES OF DRIFT-AWARE STRATEGY IN UNRELIABLE SPATIAL OBSERVATION RELATED ATTRIBUTES

Method	FOC		LTO	
	P.(%)	S.(%)	P.(%)	S.(%)
w/o D_t	72.5	41.3	58.1	37.6
SCRTrack	75.5	43.3	61.6	38.9

TABLE XI

ANALYSIS OF THE MCP TEMPORAL WINDOW LENGTH I ON THE FOC ATTRIBUTE

$I =$	1	5	10	15	20
P.(%)	67.6	71.0	75.5	74.2	73.8
S.(%)	39.8	41.1	43.3	42.9	42.5

rapid recovery and accurate relocalization. In contrast, removing $\mathcal{F}_r$ allows mismatched evidence to accumulate, causing drift. By modulating the discretization step size $\Delta_{\mathcal{F}}$ with correspondence reliability $\mathcal{F}_r$, the full RCP stabilizes long-range temporal propagation.

The proposed reliability metric also has several limitations. A visually similar distractor may produce both a concentrated correspondence distribution and a high center response, causing reliability to be overestimated. Moreover, $H_c^{\max}$ is a frame-level scalar shared by all query tokens; it cannot itself distinguish target tokens from locally confident background tokens. Future work may incorporate uncertainty calibration, cycle-consistency verification, or multihypothesis correspondence modeling.

3) MCP for Identity Preservation Without Visual Evidence: The MCP is designed to prevent identity drift caused by abrupt response-peak relocation under occlusion or distractor interference. As shown in Table V, the MCP brings notable gains on occlusion-related attributes, improving success by 4.6% on long-term occlusion (LTO) and 8.8% on full occlusion (FOC), indicating its effectiveness when visual evidence is unreliable.

TABLE XII

COMPUTATIONAL OVERHEAD OF THE PROPOSED MODULES COMPARED WITH THE BASELINE

Method	Params (M)	FLOPs (G)	Latency (ms)	FPS
Baseline	34.384	15.713	8.43	118.6
+ MFF	62.658	30.566	10.73	93.2
+ RCP	87.826	35.160	11.78	84.9
+ MCP	34.384	15.713	8.90	112.4
SCRTrack	116.100	50.013	15.13	66.1

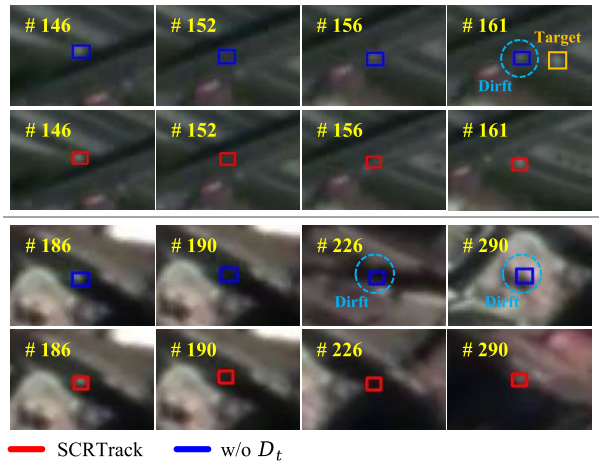


Fig. 8. Visualization results of SCRTrack (with D_t) and without D_t in occlusion scenes.

Table X further validates the MCP design. Trajectory prediction without motion consistency monitoring (w/o D_t) yields only limited improvement, while the full MCP achieves the best performance. This shows that motion prediction alone is insufficient; detecting abnormal response-peak displacement is crucial to preventing unreliable appearance-driven updates. Table XI reports the sensitivity analysis of the MCP temporal window length I. When $I = 1$, the threshold is estimated from the latest displacement, making it vulnerable to noisy response-peak fluctuations. Increasing I to 5 improves stability, but the motion baseline is still insufficiently robust. The best

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Fig. 9. Visualization results of trackers in complex scenes. (a) SatSOT/car_45 (attributes: TO, POC, SOB, BJT, and ROT). (b) OOTB/car_31 (attributes: IPR, PO, MB, BC, SA, LT, and AM). (c) SatSOT/car_07 (attributes: BC, ROT, and TO). (d) SV248S/02_000002 (attributes: DS, ND, BCL, and IPR). (e) SV248S/04_000017 (attributes: STO, DS, IV, BCH, ND, BCL, and IPR). (f) SV248S/04_000005 (attributes: STO, LTO, BCH, CO, BCL, and IPR).

performance is obtained at $I = 10$, with 75.5% precision and 43.3% success, suggesting that a short-term history of about ten frames can capture the locally smooth motion pattern of satellite targets while filtering out frame-level noise. When I is further enlarged to 15 or 20, performance decreases because the estimated threshold becomes oversmoothed and less responsive to recent abnormal peak relocation. Therefore, we adopt $I = 10$ as a tradeoff between noise suppression and motion adaptability.

Fig. 8 shows the typical occlusion cases. Without D_t, the response peak may shift to similar distractors or background regions. The MCP detects such abrupt displacement, suppresses unreliable predictions, and temporarily relies on motion-based estimation until reliable visual evidence reappears, thereby reducing identity switches and improving long-term stability.

4) Analysis of Feature Forming, Propagating, and Maintaining: The spatial and temporal unreliability are coupled in satellite videos, where weak observations may contaminate temporal states and unstable propagation can amplify localization errors. Thus, robust SVSOT requires coordinated regulation of spatial evidence and temporal state evolution. As shown in Table V, combining MFF and RCP brings larger gains than using either alone, demonstrating their complementarity: MFF provides cleaner spatial features for temporal modeling, while RCP suppresses unreliable correspondence propagation. The MCP further protects recursive states when

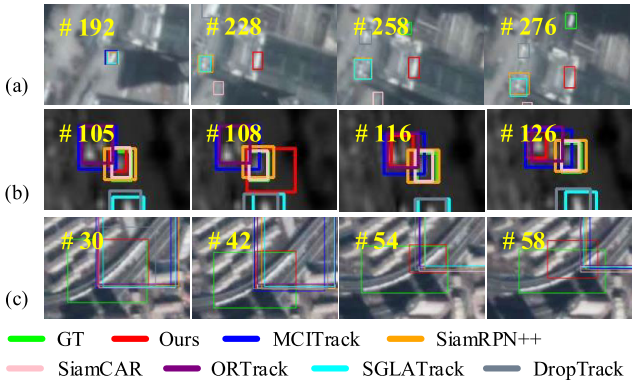


Fig. 10. Representative failure cases of SCRTrack. (a) LTO (over 50 frames) with nonlinear motion, (b) densely distributed similar targets with consistent movement patterns, and (c) aspect-ratio variation.

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localization is degraded by occlusion or distractors. These results indicate that SCRTrack is a unified reliability-aware spatiotemporal regulation framework rather than a simple module combination. By jointly regulating spatial feature formation, temporal evidence propagation, and identity-consistent state maintenance, SCRTrack suppresses error accumulation and improves robustness in challenging satellite video tracking.

5) Computational Overhead Analysis: Table XII reports the computational overhead introduced by the proposed modules. All variants are evaluated on the same hardware, with the same input size and timing protocol. The MCP introduces no additional learnable parameters and negligible neural-network FLOPs, although its displacement monitoring and Kalman filter operations incur a small runtime overhead. The proposed modules increase the computational cost, but SCRTrack still runs at 66.1 FPS and retains real-time processing capability while improving tracking robustness.

E. Visualization Under Challenging Scenarios

To qualitatively complement the quantitative results, we visualize representative tracking results of SCRTrack and several state-of-the-art trackers on challenging satellite video sequences in Fig. 9. These examples include typical SVSOT challenges, such as cluttered backgrounds, visually similar distractors, appearance variation, ambiguous correspondence, and occlusion. In Fig. 9(a) and (b), several compared trackers are attracted by distractors with similar appearance and suffer from identity switches or tracking drift. SCRTrack maintains more stable localization, benefiting from calibrated spatial evidence and identity-aware state maintenance. In Fig. 9(c) and (d), in-plane ROT and appearance changes introduce ambiguous temporal correspondence. Compared with other trackers, SCRTrack produces more consistent trajectories by suppressing unreliable temporal updates. In Fig. 9(e) and (f), occlusion causes some trackers to lose the target or drift to nearby distractors. SCRTrack preserves trajectory continuity during occlusion and relocalizes the target after reappearance through motion-consistent identity preservation. These qualitative results further show that rather than independently improving isolated components, SCRTrack jointly calibrates spatial evidence, regulates temporal propagation, and preserves identity-consistent states. Overall, the visual comparisons provide evidence for the proposed reliability-aware spatiotemporal consistency regulation in suppressing error accumulation when spatial clutter, ambiguous correspondence, and occlusion co-occur in satellite video tracking.

V. Limitations

Although SCRTrack improves robustness under unreliable satellite observations, it may still fail in extremely ambiguous cases, as shown in Fig. 10. Under LTO (over 50 frames) with nonlinear motion, MCP may become unreliable when the target trajectory deviates from its recent motion trend. Similarly, abrupt acceleration or unusually large displacement may trigger the KF branch, and if the actual motion deviates from the KF prediction, the resulting localization may drift. In scenes containing extremely densely distributed similar targets with consistent movement patterns, the response peak may switch to a nearby distractor when appearance and motion cues are both indistinguishable. For elongated targets undergoing turning-induced aspect-ratio variation, the predicted box may cover only part of the object, suggesting limited adaptability to changing target geometry. This case illustrates aspect-ratio variation, whereas robustness to pronounced scale changes remains insufficiently evaluated. Future work will explore stronger long-term trajectory reasoning, instance-aware distractor discrimination, and geometry-adaptive box estimation.

VI. Conclusion

In this article, we present SCRTrack, a spatiotemporal consistency regulation framework for robust satellite video single-object tracking. The key insight of SCRTrack is that tracking failures in satellite videos originate from the recursive accumulation of unreliable evidence. To address this issue, SCRTrack jointly calibrates spatial representations, regulates temporal state propagation, and preserves identity-consistent target states. Specifically, the MFF improves spatial evidence formation by purifying shallow fine-grained cues and selectively injecting them into deep semantic representations. The RCP estimates correspondence reliability from interframe matching distributions and uses it to adaptively regulate state-space updates, suppressing the propagation of mismatched temporal evidence. The MCP adopts motion-guided prediction when appearance-based localization becomes unreliable, reducing identity switches under occlusion and distractor interference. Extensive experiments on SatSOT, SV248S, and OOTB show that SCRTrack achieves the state-of-the-art performance under diverse satellite tracking challenges. Comprehensive attribute-based analysis and ablation studies further verify the complementary contributions of MFF, RCP, and MCP. Overall, SCRTrack provides an effective reliability-aware spatiotemporal consistency regulation paradigm for robust tracking in satellite videos.

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